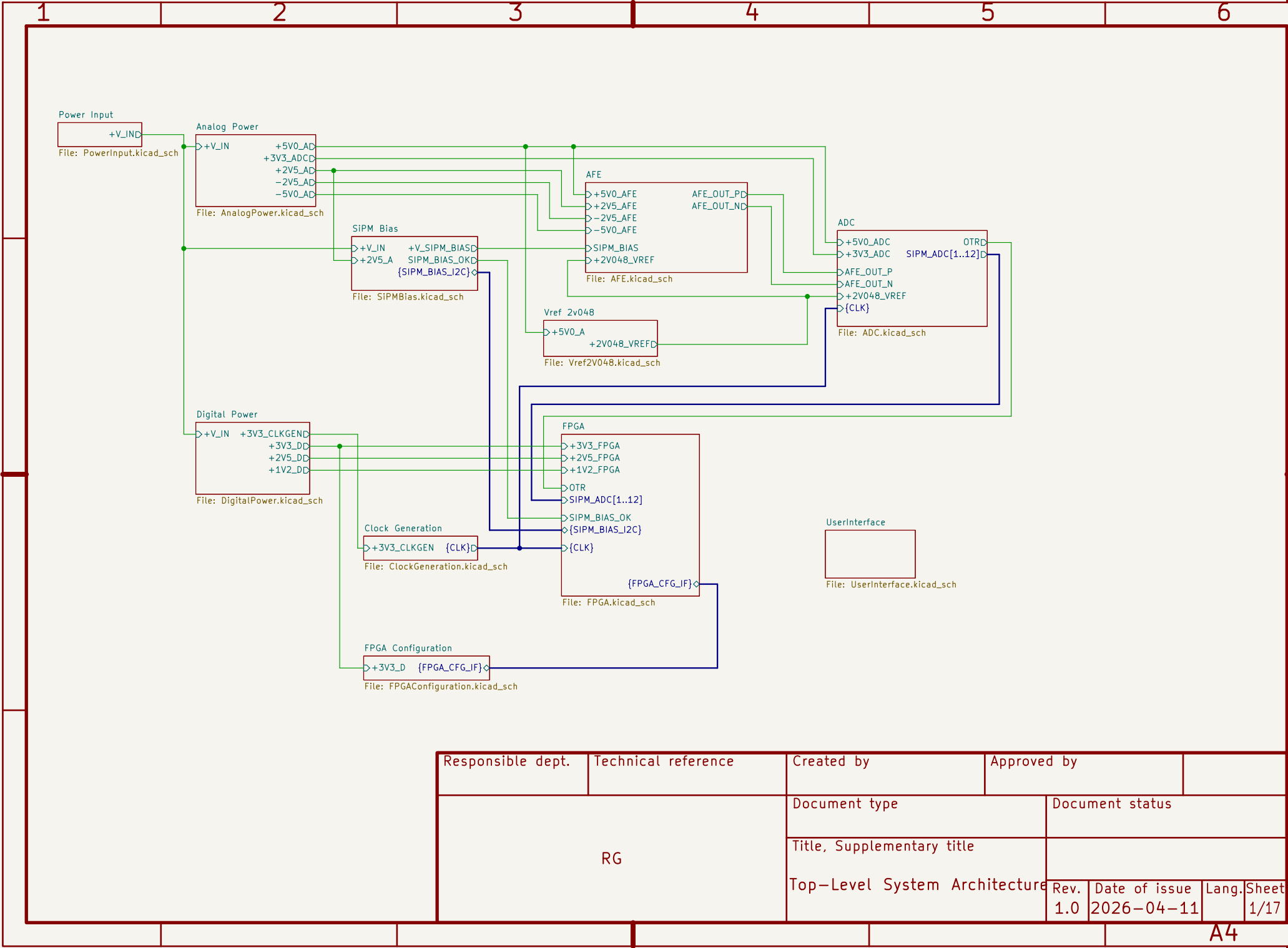
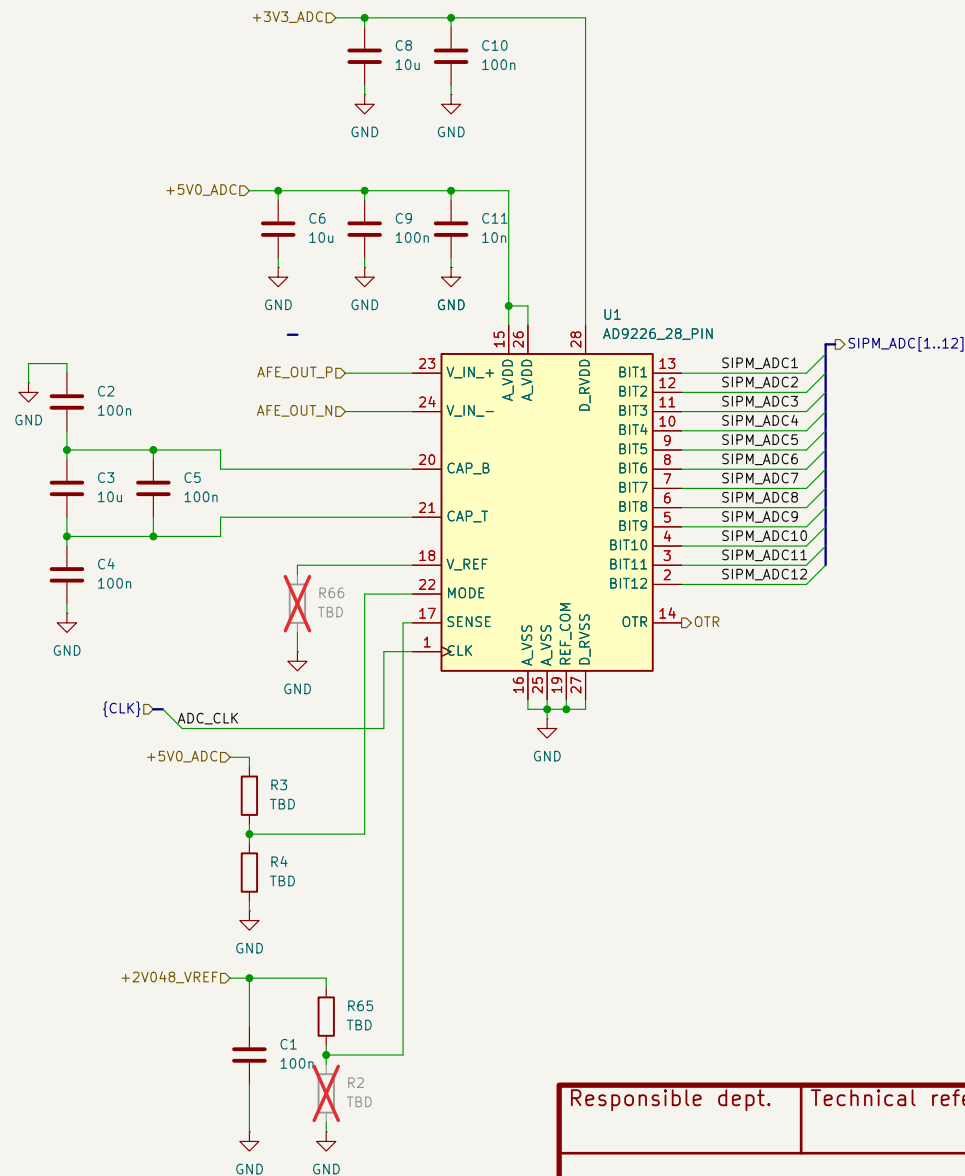
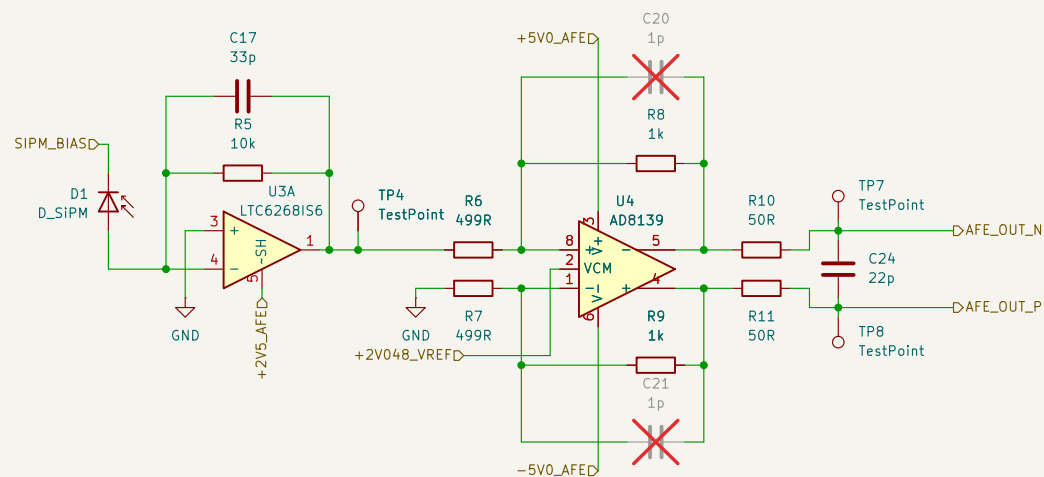




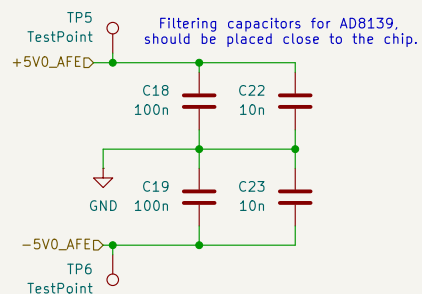
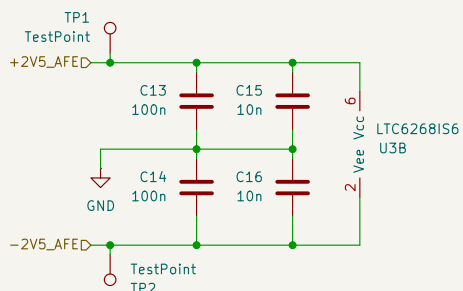


Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
		12-bit 65MSPS Analog-to-Digital Converter	Rev. 1.0	Sheet 2/17



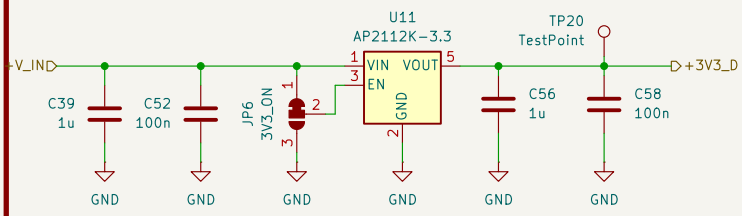
All capacititors (except for voltage filtering) are COG / NP0.
All resistors are 1% or better.

Label	Description
VCC_SIPM	28.89V–29.22V (+322.5mV over 15degC) SIPM bias
VCC_5V	Differential ADC Driver and its reference voltage
VCC_5V	Differential ADC Driver
VCC_2V5	TIA
VEE_2V5	TIA

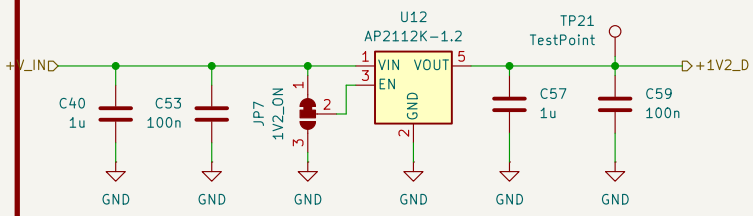


Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
		TIA & Differential ADC Driver	Rev. 1.0	Date of issue 2026-04-11
			Lang.	Sheet 3/17

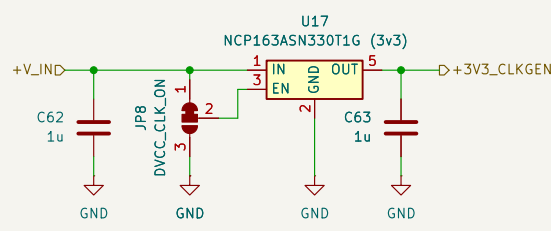
1	2	3	4	5	6	
						A
						B
						C
						D
		Responsible dept.	Technical reference	Created by	Approved by	
		RG	Document type		Document status	
			Title, Supplementary title			
			User Interface		Rev. 1.0	Date of issue 2026-04-11
					A4	



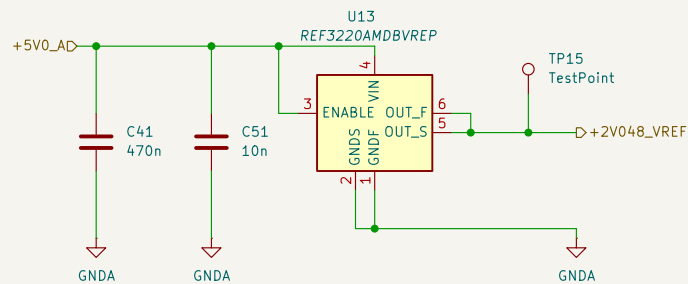
Label	Direction	Description
VCC	Input	Raw input voltage (6V)
VCC_SIPM	Output	30V for analog frontend bias
VCC_3V3	Output	3.3V for FPGA I/O, flash, ADC digital side
VCC_1V2	Output	1.2V for FPGA core
VCC_5V	Output	5V for ADC
VCC_2V5	Output	2V5 for amplifier
VEE_2V5	Output	-2V5 for amplifier
GND	Bidirectional	Ground



+2V5_D



Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
		Digital Power Regulators (3.3V, 1.2V)	Rev	
		Date of issue	Lang.	Sheet
		1.0	2026-04-11	9/17



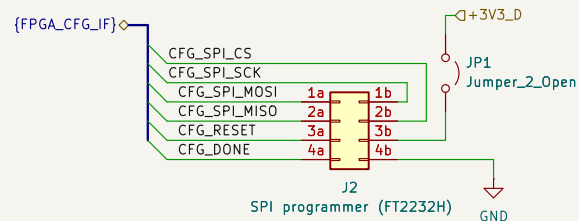
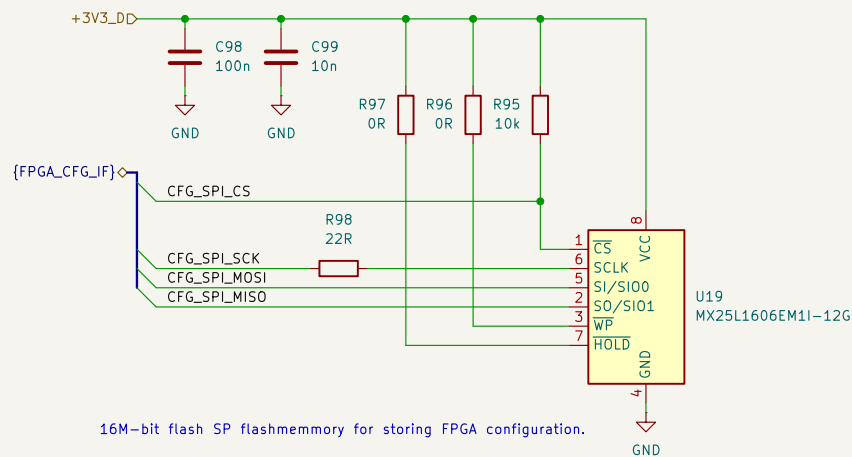
REF3220AMDBVREP is 2,048V precision voltage reference.

Place U13 (REF3220) physically close to the AD9226 ADC.

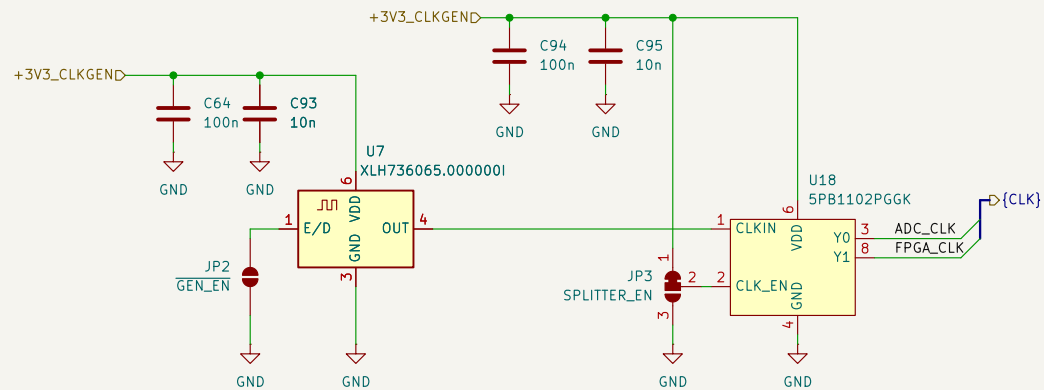
Kelvin Sense Connections:

- OUT_S (Sense+) must be routed as a dedicated trace directly to the ADC's V_REF pin.
- GND_S (Sense-) must be routed as a dedicated trace directly to the ADC's REF_COM pin.

Responsible dept.	Technical reference	Created by	Approved by			
RG		Document type		Document status		
		Title, Supplementary title				
				Rev. 1.0	Date of issue 2026-04-11	Lang.

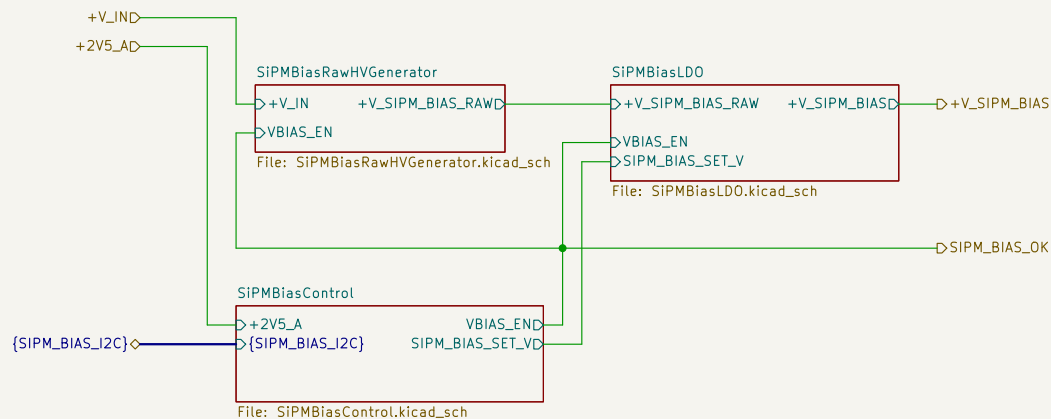


Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
FPGA Configuration Memory – External 16Mbit SPI Boot-Flash		Rev. 1.0	Date of Issue 2026-04-11	Lang. Sheet 12/17

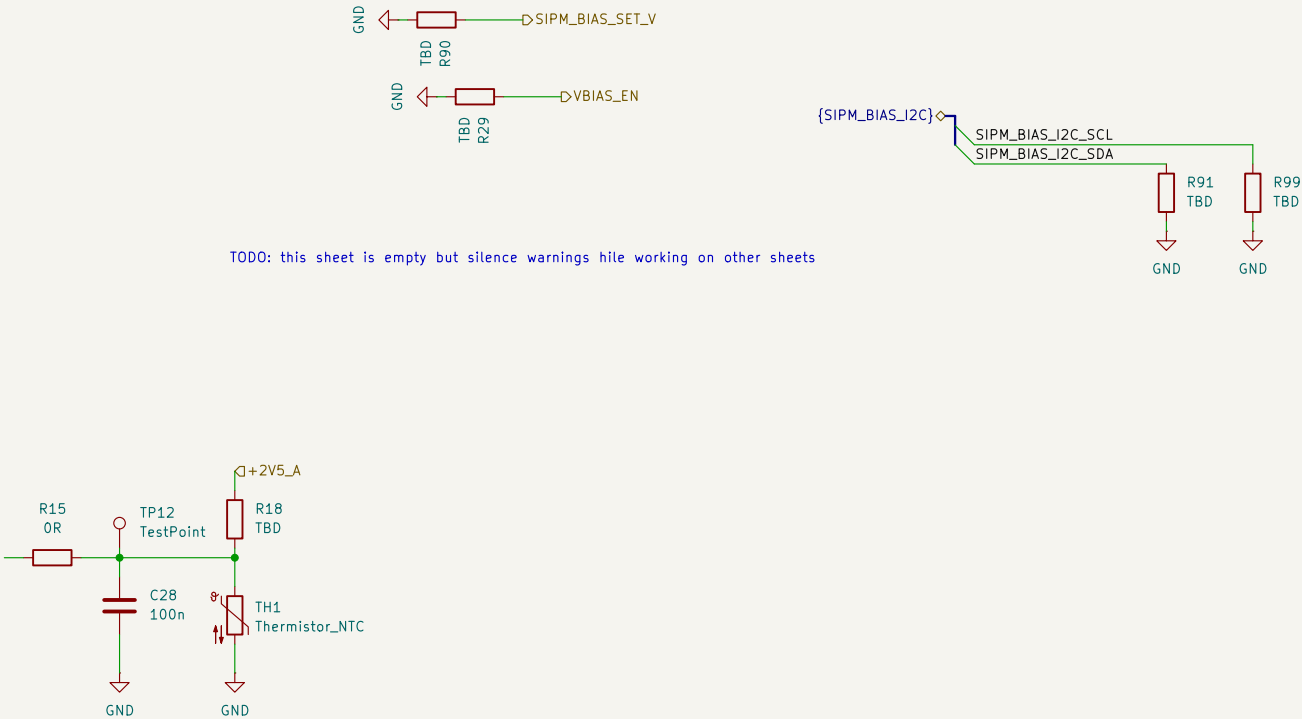


CLK is 65MHz 25ppm
5PB1102PGGK is 1:2 CLK splitter

Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
System Clock – 65 MHz Oscillator and Dual Output Buffer		Rev. 1.0	Date of issue 2026-04-11	Lang. Sheet 13/17

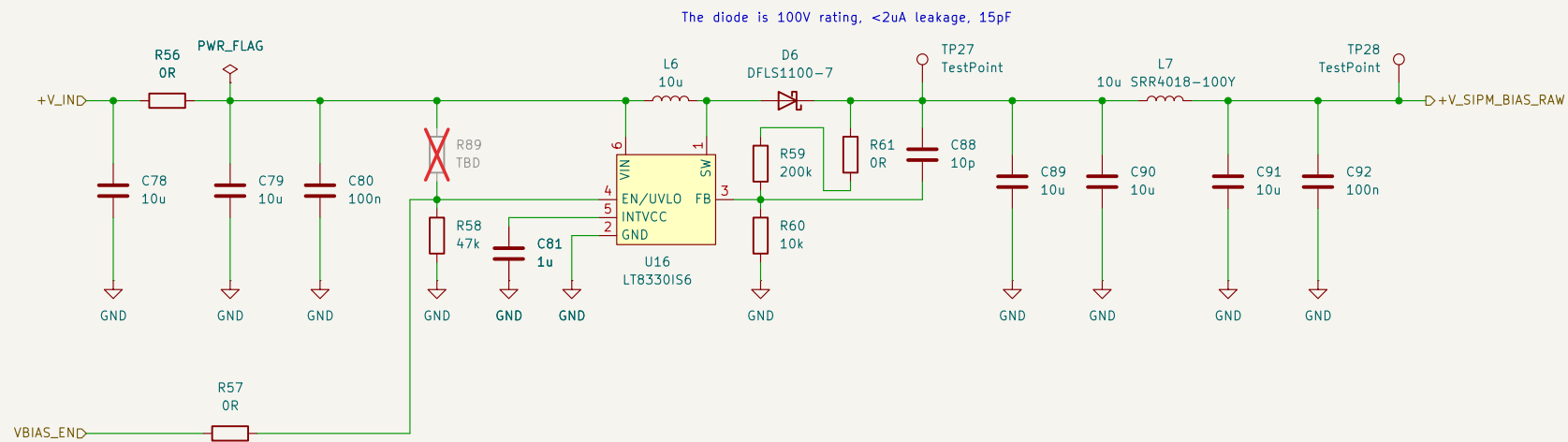


Responsible dept.	Technical reference	Created by	Approved by				
RG		Document type		Document status			
		Title, Supplementary title					
				Rev.	Date of issue	Lang.	Sheet
				1.0	2026-04-11		15/17

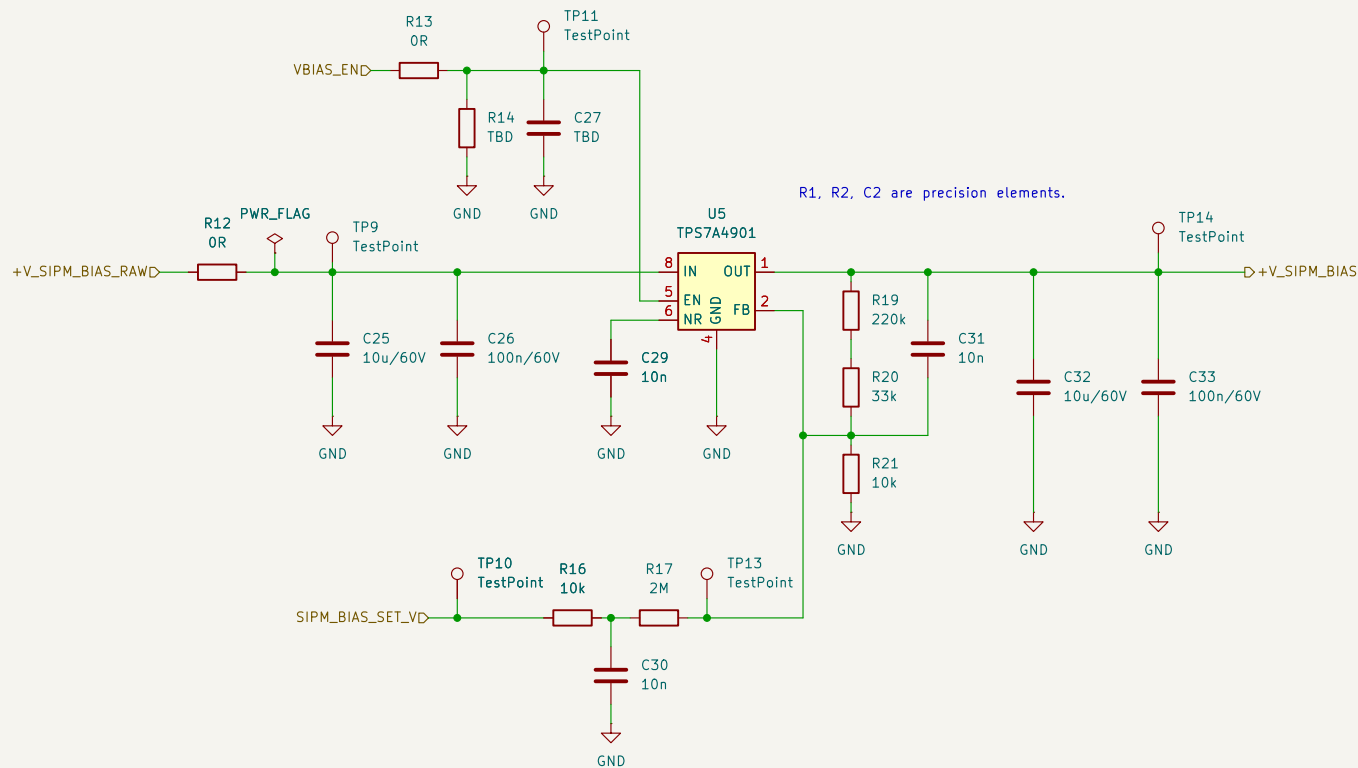


TODO: this sheet is empty but silence warnings hile working on other sheets

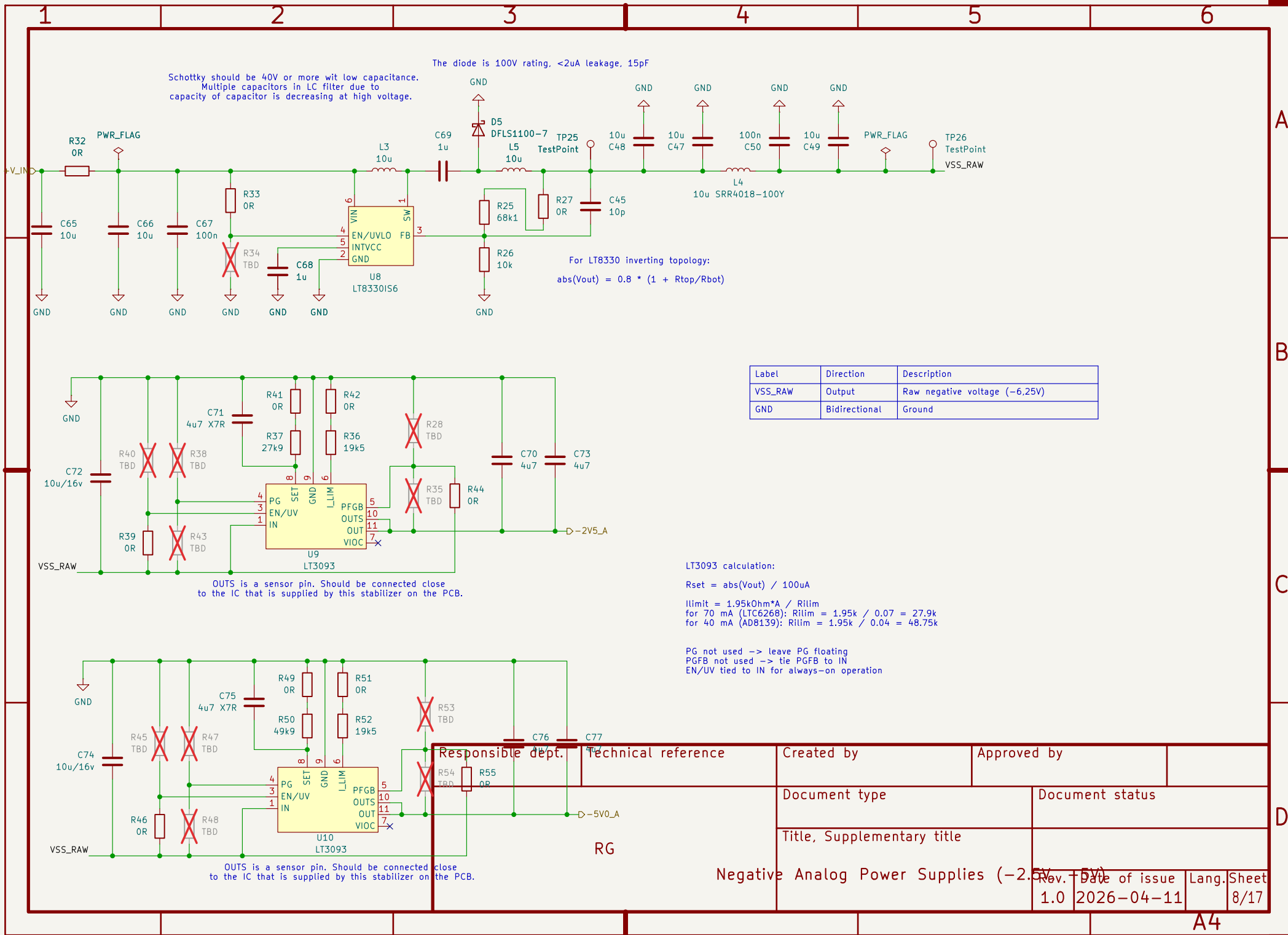
Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
SiPM Bias Control and Temperature Compensation		Rev.	Date of issue	Lang.
		1.0	2026-04-11	
		Sheet	4/17	

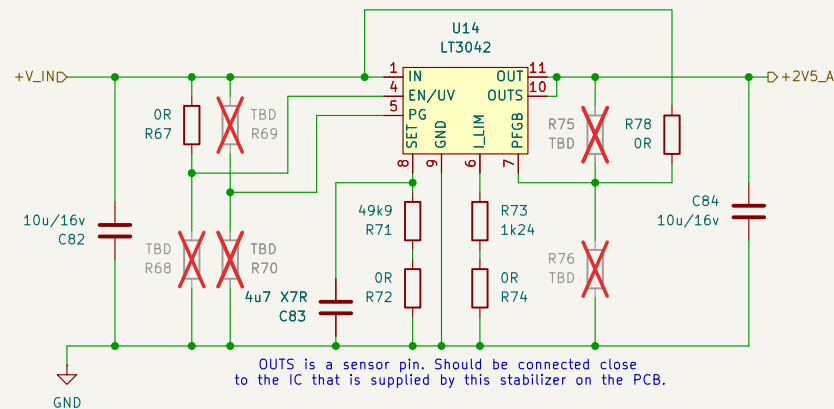


Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type	Document status	
		Title, Supplementary title		
		High-Voltage SiPM Bias Generator & Ultra-Low Noise LDO	Rev. 1.0	Sheet 5/17
		Date of Issue 2026-04-11	Lang.	



Responsible dept.	Technical reference	Created by	Approved by			
RG		Document type		Document status		
		Title, Supplementary title				
				Rev. 1.0	Date of issue 2026-04-11	Lang.





LT3042 calculation:

$$R_{set} = \frac{\text{abs}(V_{out})}{100\mu A}$$

$$I_{limit} = \frac{125\text{mA} \cdot k\Omega\text{m}}{R_{lim}}$$

For:

$$\text{for } 60 \text{ mA: } R_{lim} = \frac{125}{60} = 2.08\text{k}$$

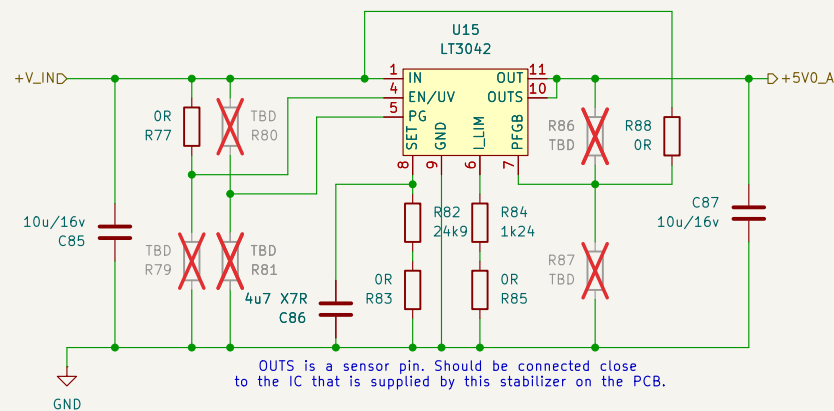
$$\text{for } 70 \text{ mA: } R_{lim} = \frac{125}{70} = 1.79\text{k}$$

$$\text{for } 100 \text{ mA: } R_{lim} = \frac{125}{100} = 1.24\text{k}$$

PG not used -> leave PG floating

PGFB not used -> tie PGFB to IN

EN/UV tied to IN for always-on operation



HIGH QUALITY CAPS FOR SET PIN!!!

Responsible dept.	Technical reference	Created by	Approved by	
RG		Document type		Document status
		Title, Supplementary title		
		Positive Analog Power Supplies (2.5V - 5V)		Rev. 1.0 Date of issue 2026-04-11 Lang. Sheet 10/17